

PASSENGERS MONITORING SYSTEM

Community Service Project Report

Submitted to the Faculty of

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
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In partial fulfilment of the requirements for the award of the Degree of

**BACHELOR OF TECHNOLOGY
IN
Artificial Intelligence and Data Science**



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CERTIFICATE

This is to certify that the community service project report entitled “**PASSENGERS MONITORING SYSTEM**” is a bonafide record of work carried out by **R. Sai Nithin (20481A5448), A. Khadar Jilani (20481A5401), V. Kondala Rao (20481A5461), D. Soma Venkat (20481A5413)** under the guidance and supervision of **Mr. K. Ashok Reddy M.Tech,(PhD)** partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Artificial Intelligence and Data Science** of **Seshadri Rao Gudlavalleru Engineering College, Gudlavalleru.**

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ABSTRACT

The use of Public Transportation can be increased by the developing the Technology in the Transportation system. In the smart world the technology must very useful to live the life of the huma being in a peaceful way. The main problem faced by the authorities of transportation department and passengers is not knowing the availability of the seats and number of passengers using the Transportation facility. For the above problem, the simple solution is to develop the passenger monitoring system in the public Transportation, which will be particularly useful to the both the authorities and passengers. Using this IOT based device, it is user friendly and has exceptional features like showing availability of seats, number of passengers are on the bus etc....This system provides better technology not only for the Bus Transportation System but to all other Public Transportations. We basically take the reference of the Bus Transportation to show the abstract of the IOT device.

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CHAPTER-I

INTRODUCTION

1.1 Scope of the Project

In our country one of the most problem is organization of public Transportation system. There typical transportation system where includes various corruption and also added that public do not get their enough good transport system for their comfort. We also find that the owner of vehicle does not get his proper profit due to some corrupted employee.

Passenger counting and detection is an important task for the traffic monitoring and utilization of resources for a public transport system. It is important to monitor public traffic efficiently for a well-organized and cost-effective public transport system. Public transport companies provide information on the connecting routes which plays an essential role in the traffic monitoring system. The allocation of public vehicles to the various connection routes during normal and busy hours are essential for the management of a transport system. These connecting routes vary from time to time in different places. Statistical tests may require for getting a better estimation of the average number of vehicles needed for a particular route. Traffic overload is more significant in the daytime as compared to night. More public vehicles are required on a route during busy hours. The correct distribution of available resources or vehicles over different routes is essential for better optimization and management of a transport network. This requires continuous monitoring of the transport system such as counting the passengers during getting off and on the public transport. The analysis of this data to count the passengers can be carried out using a computer vision-based approach to avoid use of human efforts.

1.2 OBJECTIVES OF THE PROJECT

The main Objectives of the Passengers Monitoring System are

- Passengers Count in the bus at that Time.
- Increase in user satisfaction.
- Increase in the number of passengers by knowing the count.
- Reduction of traffic congestions and greenhouse emissions by reducing the use of unwanted use of transportation.
- Significant cost savings in labour-intensive manual procedures.
- Respect privacy concerns.
- Utilize the maximum capacity of a passenger vehicle.
- Obtain information for forecasting and service-related decisions.
- By knowing how passengers are distributed throughout a vehicle, companies can optimize the layout to improve comfort and safety.
- Analyse key performance indicators like passengers per mile and cost per passenger.
- Record times spent waiting at each stop.
- Monitor passenger numbers over time.
- Justify grants.
- Reliable data at a low cost, regardless of how busy the service is.
- Monitor live occupancy to follow social distancing and ensure safety.
- Increase resource availability.
- Route optimization.

1.4 PROBLEM IDENTIFIED

There are several problems that can affect public transportation, including overcrowding, inadequate funding, and inefficient routes. These problems can lead to delays, increased travel times, and reduced reliability for passengers. In some cases, public transportation may not be accessible to certain individuals, such as those with disabilities, which can limit their ability to access certain services or destinations. Additionally, the lack of investment in public transportation infrastructure can lead to outdated and poorly-maintained vehicles and facilities, which can be uncomfortable and unsafe for passengers.

The main problem faced by the authorities of transportation department and passengers is not knowing the availability of the seats and number of passengers using the Transportation facility. The typical transportation system where includes various corruption and also added that public do not get their enough good transport system for their comfort.



CHAPTER-II

LITERATURE SURVEY

2.1 ANALYSIS OF THE PROBLEM

One possible analysis of problems related to crowding on buses could involve examining data on bus ridership and capacity on a particular route or system. For example, the analysis might look at the number of passengers boarding buses at different times of day, and compare this to the capacity of the buses on the route. If the data shows that buses are consistently operating above capacity, particularly during peak times, this could indicate a problem with crowding.

The analysis could then consider the potential causes of this crowding, such as an insufficient number of buses on the route, inadequate scheduling, or high levels of demand for bus service. It could also look at the effects of crowding on passengers, such as longer wait times at bus stops, discomfort and inconvenience on crowded buses, and potential safety concerns.

Based on this information, the analysis could propose potential solutions to the problem of crowding on buses. These could include increasing the frequency of bus service on the route, implementing dynamic routing to adjust to changes in demand, or providing alternative modes of transportation, such as shuttle buses or ride-sharing services. The analysis could also recommend further data collection and analysis to better understand the problem and evaluate the effectiveness of potential solutions.

2.2 EXISTING TECHNIQUES

1. [1] Edge analysis methods have been used for tracking and detection of objects. The edge follows a particular shape. For example, human head corresponds to a circular shape. An edge filter, is applied to the object to extract edge information. Edge detection is important for the extraction of the skeleton of the silhouette. The main problem related to skeleton computation is its sensitivity to noise, although the noise from silhouettes can be reduced using smoothing filters.
2. [2] Spatiotemporal methods are based on the selection of a line from the region of interest. The statistical model techniques are applied to estimate the number of passengers crossing the line. Although these techniques are fast, simple and easy to implement, these algorithms fail (in some cases) to provide accurate results for passengers counting, as the foreground generated from a stationary person is considered as multiple detected people in these techniques.
3. [3] Motion detection-based techniques follow two steps. Firstly, moving regions of objects and humans are detected in a video. Information about the trajectories and direction of a moving person are provided in the next step. These techniques recognize and count people who cross a virtual line. Trajectory and clustering algorithms can count passengers. Features of passengers are extracted and passenger direction and estimations are monitored. Due to complex environmental conditions (such as overlapping among passengers), the position of trajectories becomes complex and the accuracy of clustering is affected. In addition, the detection system counts two persons as a single person when two or more passengers follow the same trajectory.

CHAPTER-III

PROPOSED METHOD

3.1 METHODOLOGY

In this project, we will make a Bidirectional Passengers Counter with an Automatic Light Control as a proto type System using Arduino. This project is based on a pair of Infrared Sensor that detects interrupt when it detects an obstacle. The pair of IR sensors can detect the visitor from both directions, i.e., the number of entering passengers and the number of exiting passengers.



This Arduino Bidirectional Passengers Counter Project can be used to count the number of persons entering in to the bus or any vehicle which is placed in the entrance gate. It can also be used at gates of parking areas and other public places. The device counts the total number of people entering through the gate and the total number of people leaving through the same gate. And finally, it counts the total number of people currently present inside the room. When no people are inside the room, i.e., the total number of people is zero then the shows turning of the light which in used as proto type. When even a single person is found inside the bus, the light turns on. The light control system is automatic based on the passenger's presence. We can make the entire project using a single Arduino Nano Board. We can use a 16x2 LCD Display to show the number of passengers. But you can prefer a 0.96" OLED Display for the project. A 5V Single channel relay gets activated when a person is detected inside a bus. The light turns ON automatically which is connected to the Relay.

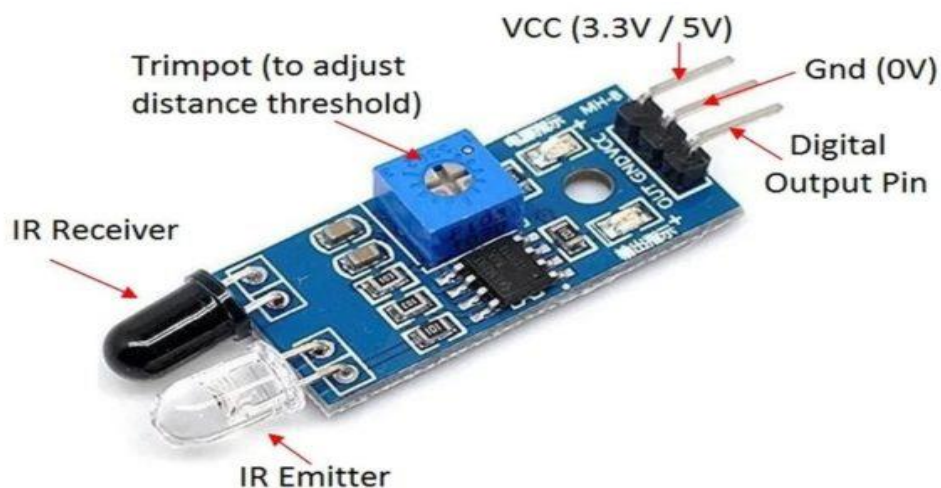
3.2 COMPONENTS REQUIRED

- Infrared Sensor
- Arduino Uno Board
- OLED Display
- Relay Module
- Power Supply
- Connecting Wires

3.2.1 INFRARED SENSOR

The main element of this project is the IR Sensor which works as a Human Detector. Whenever the IR sensor detects an interrupt, it counts the person and adds it to the previous value. IR Sensor module has the great adaptive capability of the ambient light. It has an infrared transmitter and a receiver. The infrared emitting tube emits a certain frequency which when encounters an obstacle reflect back to the signal. The reflected signal is then received by the receiver tube. Apart from the IR Transmitter and Receiver the circuit also consists of Opamp, Variable Resistor (Trimmer pot) & output LED.

The Sensor consists of the following Parts.



1. IR LED Transmitter

IR LED emits light, in the range of Infrared frequency with a wavelength of 700nm – 1mm. IR LEDs have a light-emitting angle of approx. 20-60 degrees and has a range up to 5-10cm. IR LED white or transparent in colour, so it can emit the maximum amount of light.

2. Photodiode Receiver

Photodiode acts as the IR receiver as it conducts when light falls on it. Photodiode looks like a LED, with a black colour coating on its outer side. The Black colour absorbs the highest amount of light.

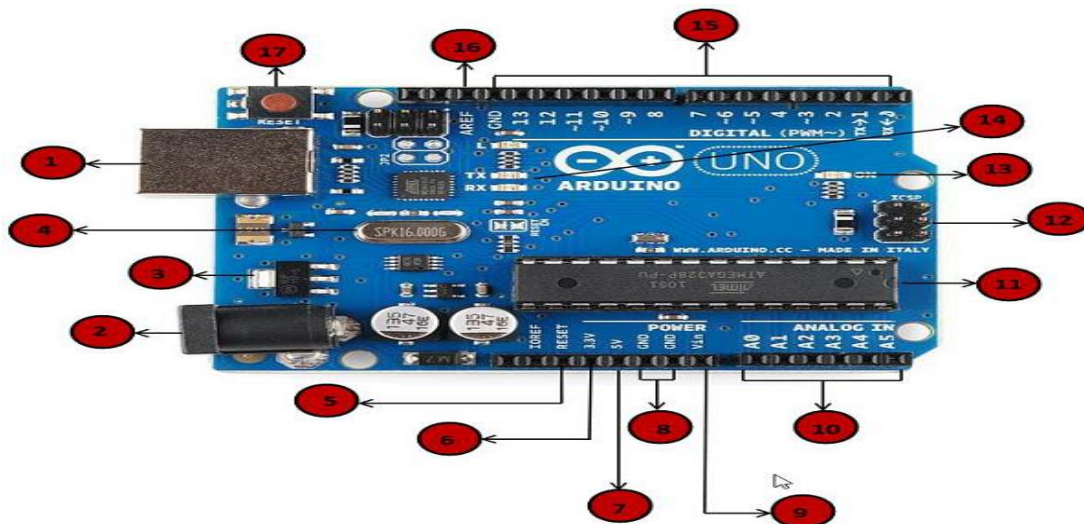
3. LM358 Opamp

LM358 is an Operational Amplifier (Op-Amp) is used as a voltage comparator in the IR sensor. The comparator circuit compares the threshold voltage set using the pre-set and the photodiode's series resistor voltage. When the Photodiode's series resistor voltage drop is greater than the threshold voltage, the Op-Amp output is high. Similarly, when the Photodiode's series resistor voltage drop is lesser than the threshold voltage, the Op-Amp output is low. When the Op-Amp output is high the LED at the output terminal turns ON. This indicates an object or obstacle has been detected.

4. Variable Resistor

The variable resistor used here is pre-set. It is used to calibrate the distance range at which the object should be detected.

3.2.2 ARDUNIO UNO



1. Power USB

Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection (1).

2.Power (Barrel Jack)

Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel Jack (2).

3.Voltage Regulator

The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.

4.Crystal Oscillator

The crystal oscillator helps Arduino in dealing with time issues. By using the crystal oscillator. The number printed on top of the Arduino crystal is 16.000H9H. It tells us that the frequency is 16,000,000 Hertz or 16Mhz.

5,17. Arduino Reset

You can reset your Arduino board, i.e., start your program from the beginning. You can reset the UNO board in two ways. First, by using the reset button (17) on the board. Second, you can connect an external reset button to the Arduino pin labelled RESET.

6,7,8,9. Pins (3.3, 5, GND, Vin)

- 3.3V (6) – Supply 3.3 output volt
- 5V (7) – Supply 5 output volt
- Most of the components used with Arduino board works fine with 3.3 volt and 5 volt.
- GND (8)(Ground) – There are several GND pins on the Arduino, any of which can be used to ground your circuit.
- Vin (9) – This pin also can be used to power the Arduino board from an external power source, like AC mains power supply.

10.Analog pins

The Arduino UNO board has six analog input pins A0 through A5. These pins can read the signal from an analog sensor like the humidity sensor or temperature sensor and convert it into a digital value that can be read by the microprocessor.

11.Main microcontroller

Each Arduino board has its own microcontroller (11). You can assume it as the brain of your board. The main IC (integrated circuit) on the Arduino is slightly different from board to board. The microcontrollers are usually of the ATMEL Company. You must know what IC

your board has before loading up a new program from the Arduino IDE. This information is available on the top of the IC.

12.ICSP pin

Mostly, ICSP (12) is an AVR, a tiny programming header for the Arduino consisting of MOSI, MISO, SCK, RESET, VCC, and GND. It is often referred to as an SPI (Serial Peripheral Interface), which could be considered as an "expansion" of the output. Actually, you are slaving the output device to the master of the SPI bus.

13.Power LED indicator

This LED should light up when you plug your Arduino into a power source to indicate that your board is powered up correctly. If this light does not turn on, then there is something wrong with the connection.

14.TX and RX LEDs

On your board, you will find two labels: TX (transmit) and RX (receive). They appear in two places on the Arduino UNO board. First, at the digital pins 0 and 1, to indicate the pins responsible for serial communication. Second, the TX and RX led (13). The TX led flashes with different speed while sending the serial data. The speed of flashing depends on the baud rate used by the board. RX flashes during the receiving process.

15.Digital I/O

The Arduino UNO board has 14 digital I/O pins (15) (of which 6 provide PWM (Pulse Width Modulation) output. These pins can be configured to work as input digital pins to read logic values (0 or 1) or as digital output pins to drive different modules like LEDs, relays, etc. The pins labelled “~” can be used to generate PWM.

16.AREF

AREF stands for Analog Reference. It is sometimes, used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

3.2.3 POWER SUPPLY 5V Supply

The power supply to micro controller is given through a battery of 5 volts.

The battery is used to supply power to all microcontroller interfaced sensors.



3.2.4. 0.96" I2C OLED Display

The most attractive thing about using the OLED displays is that they do not need a back-light like conventional LCD/LED screens. The organic material itself has a property known as Electroluminescence (EL), which causes the material to “glow” when stimulated by a current or an electric field. Best energy saving displays ever.

This 0.96" OLED Display Module offers 128×64pixel resolution. They are featuring much less thickness than LCD displays with good brightness and also produce better and true colours. It produces blue text on black background with very good contrast when supplied with DC 2.8V supply. The OLED Display Modules also offers a very wide viewing angle of about greater than 160°.



Properties:

1. Driver IC: SSD1306
2. Resolution: 128 x 64
3. Visual Angle: >160°
4. Input Voltage: 3.3V ~ 6V
5. Only Need 2 I/O Port Control
6. Pixel Color: Blue

3.2.5 RELAY MODULE

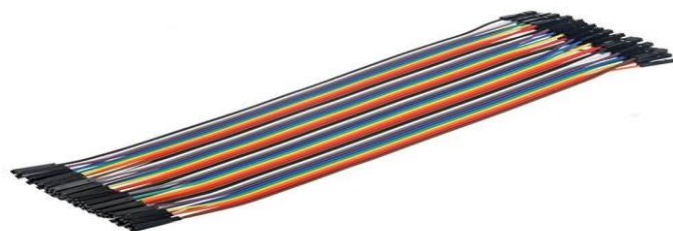


Relay is an electromechanical device that uses an electric current to open or close the contacts of a switch. The single-channel relay module is much more than just a plain relay, it comprises of components that make switching and connection easier and act as indicators to show if the module is powered and if the relay is active or not.

Pin Number	Pin Name	Description
1	Relay Trigger	Input to activate the relay
2	Ground	0V reference
3	VCC	Supply input for powering the relay coil
4	Normally Open	Normally open terminal of the relay
5	Common	Common terminal of the relay
6	Normally Closed	Normally closed contact of the relay

3.2.6 CONNECTING WIRES (JUMPER WIRES)

A Jump wire (also known as jumper, jumper wire, DuPont wire) is an electrical wire, or group of them in a cable, with a connector or pin at each end (or sometimes without them – simply "tinned"), which is normally used to interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering.



3.3 IMPLEMENTATION

The open-source Arduino Software (IDE) is required to make it easy to write and upload it to the board. For the prototype Thinker card website is used.

Implementation code:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

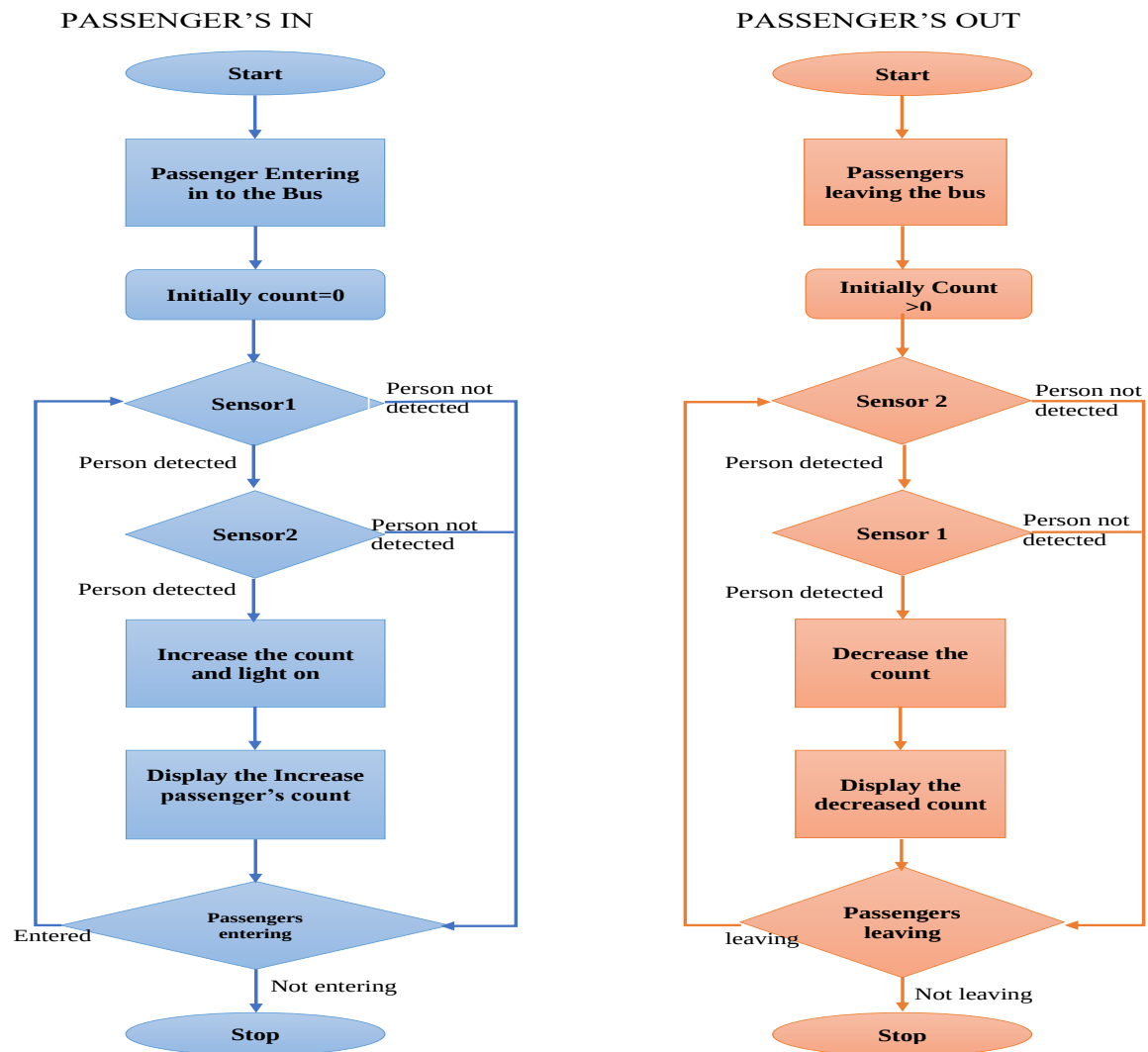
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels
#define OLED_RESET -1 // Reset pin # (or -1 if sharing Arduino reset pin)
Adafruit_SSD1306 display(SCREEN_WIDTH,
SCREEN_HEIGHT, &Wire, OLED_RESET);
#define inSensor 2
#define outSensor 3
int inStatus ,outStatus ,countin=0 ,countout=0 ,in ,out, now ;
#define relay 5
void setup()
{
    display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
    //initialize with the I2C addr 0x3C (128x64)
    delay(2000);
    pinMode(inSensor, INPUT);
    pinMode(outSensor, INPUT);
    pinMode(relay, OUTPUT);
    digitalWrite(relay, HIGH);
    display.clearDisplay();
    display.setTextSize(2);
    display.setTextColor(WHITE);
    display.setCursor(20, 20);
    display.print("Visitor");
    display.setCursor(20, 40);
    display.print("Counter");
    display.display();
    delay(3000);
}
void loop()
{
    inStatus = digitalRead(inSensor);
    outStatus = digitalRead(outSensor);
    if (inStatus == 0)
    {
```

```

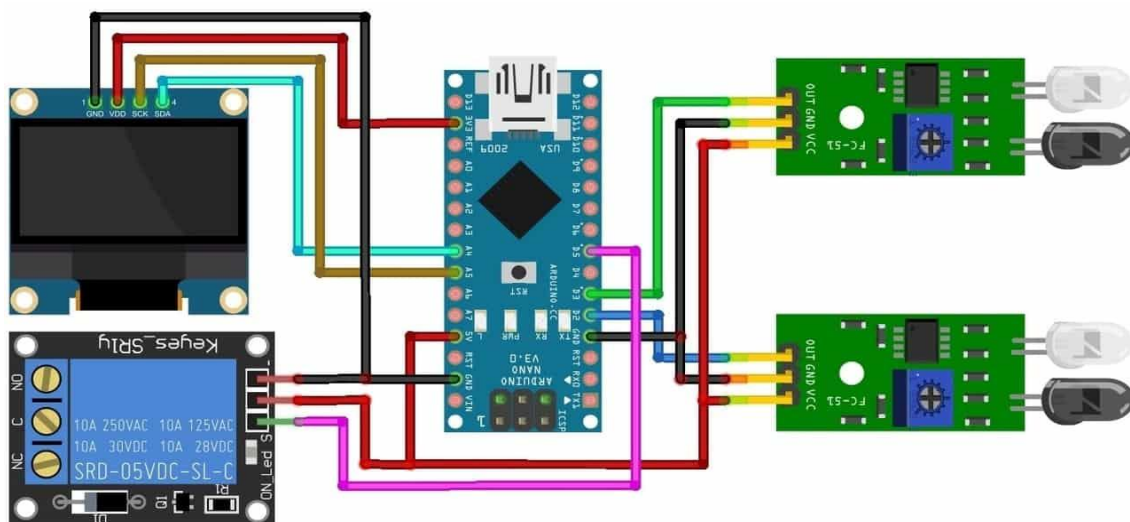
        in = countin++;
    }
    if (outStatus == 0)
    {
        out = countout++;
    }
    now = in - out;
    if (now <= 0)
    {
        digitalWrite(relay, HIGH);
        display.clearDisplay();
        display.setTextSize(2);
        display.setTextColor(WHITE);
        display.setCursor(0, 15);
        display.print("No Visitor");
        display.setCursor(5, 40);
        display.print("Light Off");
        display.display();
        delay(500);
    }
    else
    {
        digitalWrite(relay, LOW);
        display.clearDisplay();
        display.setTextColor(WHITE);
        display.setTextSize(1);
        display.setCursor(15, 0);
        display.print("Current Visitor");
        display.setTextSize(2);
        display.setCursor(50, 15);
        display.print(now);
        display.setTextSize(1);
        display.setCursor(0, 40);
        display.print("IN: ");
        display.print(in);
        display.setTextSize(1);
        display.setCursor(70, 40);
        display.print("OUT: ");
        display.print(out);
        display.display();
        delay(500);
    }
}

```

3.4 FLOW CHART



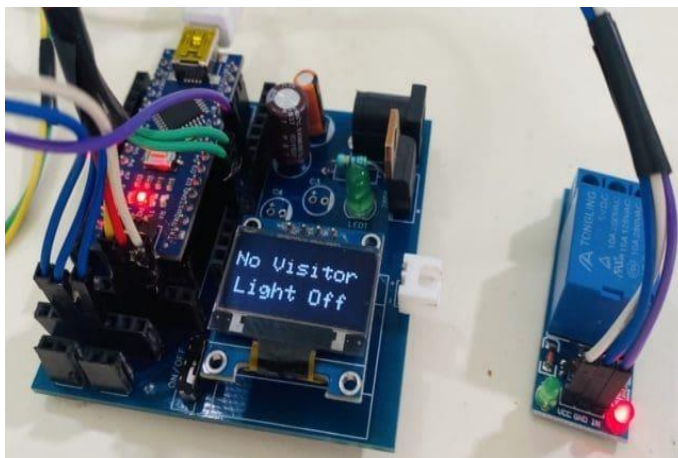
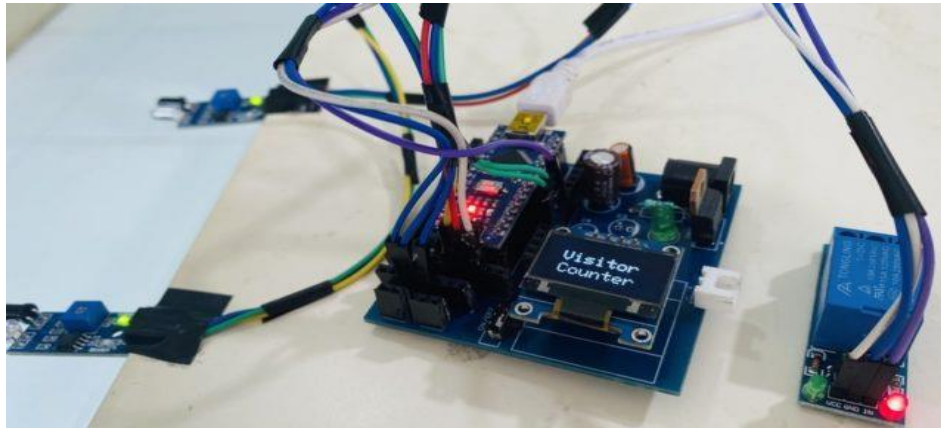
3.5 CIRCUIT DIAGRAM



CHAPTER-IV

RESULTS AND DISCUSSION

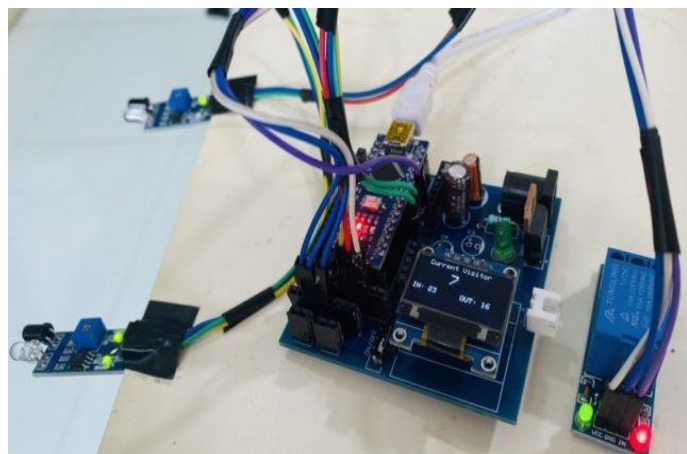
After uploading the visitor counter code to Arduino Board, the device is ready for installation. You can use a 5V DC Adapter to Power on the Device. The device has a pair of IR Sensor module. One of the IR Sensors needs to be placed at the entrance of the bus, i.e., inside top of the bus door and outside top of the bus door.



When no passengers inside the bus, the light turns off and the OLED Display will indicate no visitors are present inside the bus.

When someone makes an entry, the visitor is added and OLED Display, displays the number of incoming visitors. At this instance, the light automatically turns ON.

This is how a **Passenger Monitoring System** works.



CHAPTER-V

COMMUNITY AWARENESS PROGRAMMES CONDUCTED

5.1 FIELD SURVEY

A FIELD SURVEY WAS CONDUCTED BY OUR TEAM NEAR OUR LOCALITY IN THE TRANSPORTATION DEPARTMENT.

1. What is the capacity of normal bus?

APSRTC operates various services for passengers as follows:

- Express --50
- Super Luxury (Non-Ac, 2 + 2 Push Back) -- 36
- Ultra-Deluxe (Non-Ac, 2+2 Push Back) --40
- Indra (A.C. Seater) --40
- Amaravati (Volvo / Scania A.C Multi Axle) – 49
- Saptagiri Express -- 46
- Garuda (Volvo / Corona / Isuzu A.C. Semi Sleeper) --41
- Night Rider (Seater Cum Sleeper) – (Lower =22+5 && Upper=15)
- Vennela (2+1 A.C. Sleeper) – Lower + Upper =15+15
- Vennela (2+2 A.C. Sleeper) – Lower + Upper =12+12
- Metro Luxury A/C – 36+ 10 (Standing)
- Super Luxury Ac (2+2 Push Back) -- 36
- Pallevelugu

2. What is the condition of bus according to the passengers count?

The Condition was normal if the total seats are filled but there is some difficulty when there is standing of passengers. Sometimes we didn't know that to get the passengers without knowing the availability of seats at that time or request them to get the next bus.

3. Any problems are facing when the time of counting and ticketing?

We actually count the left behind seats when the bus is not full. And we know the actual number of seats and we analyse the count. But at the time of bus is filled it is very difficult to count the number of passengers.

4. What type of problems are you facing with authorities without knowing the count?

At the time of submission of days total amount, we are asked to tell the total number of passengers travelled in that day but it is totally different from ticket count sometimes it is

ok. But some issues will arise with the department at that time if passengers count and ticket count was not equal.

5. Did you observe that any passengers are facing problems without knowing the available seat count?

Yes, Passengers face some problems at the time of counting if the bus is completely filled with standing. Passengers also didn't know the available seats of that bus at that boarding.

6. How much time is required to count manually if there more number of passengers?

If the bus is not filled it is easy to count in the running bus. But if there is complete standing it is difficult to count in the running bus. So, there must be some halt for the journey to count the passengers after the tickets are issued.

7. If a passenger's monitoring system was developed is there any use for public transportation?

Yes, it is use full in the development of Public Transportation. In which it has advantages for both the Passengers and Employees of the department to make their work easy.

8. What is your opinion in this monitoring system?

In my opinion makes the Conductors work easy in ticketing system and he can instruct the driver to stop at the next halt point to get in the passengers or let them to take the next bus.

9. Can you give any suggestions about the model of the passengers monitoring system?

The counting device should be placed at the inside of the door and display must be visible to both inside and outside of the bus for all the passengers. It should also display the available seats. The counting must be accurate and passengers must satisfy with the System.

10. Advantages/disadvantages of passengers monitoring system in your opinion.

- Increase collected fare revenue. The increase in fare revenue produced by cross-checking passenger numbers with issued tickets can more than pay for the system.
- Save on training and employing people to manually count passengers.
- Obtain information for scheduling, forecasting and service-related decisions.
- Analyse key performance indicators like passengers per mile and cost per passenger.
- Record times spent waiting at each stop.
- Monitor passenger numbers over time.
- Reliable data at a low cost, no matter how busy the service.
- See live occupancy and help keep within social distancing.

CHAPTER-VI

INTERVENTIONS AND SERVICE PROGRAMMES TAKEN UP

1. Our team visited to the Gudivada Bus stand and conduct a service program by enquiring the problems faces by the Bus Drivers, Conductors, Employees of the Bus Depo, Passengers present in the bus stand.



2. We visited a private bus travel in Gudivada which is SKS Travels. we meet the manager and the bus drivers of the travels.



3. We also visited our college i.e., Gudlavalleru Engineering College Transportation department and conducted an survive.



CHAPTER-VII

SHORT TERM AND LONG-TERM ACTION PLANS

A short-term action plan for a passenger monitoring system project might include the following steps:

1. Identifying the key issues faced by the system and the community by this project.
2. Develop a prototype of the system and conduct initial testing to identify any potential issues or challenges.
3. Based on the results of the initial testing, refine the system and conduct further testing to ensure that it meets the project goals and objectives.
4. Identify the key stakeholders who will be affected by the project and engage with them to gather feedback and ensure their support.
5. Once the system has been tested and refined, implement it in a pilot project to gather further feedback and refine the system before rolling it out more widely.

A long-term action plan for the same project might include the following steps:

1. Develop a strategic plan for the project that outlines the long-term goals and objectives, including the key metrics that will be used to measure success.
2. Monitor the progress of the project on an ongoing basis, using the key metrics identified in the strategic plan to track progress and make adjustments as needed.
3. Identify the key challenges and opportunities that the project may face over the longer term, and develop strategies to address them.
4. Continually refine and improve the system based on feedback from stakeholders and users, and incorporate new technologies and best practices as they emerge.

CHAPTER-VIII

RECOMMENDATIONS & CONCLUSION

8.1 The suggested Recommendations are:

- Using the public transportation system in correct manner by following the rules and regulations.
- Following the perfect manner i.e., Queue system in the busses and bus stop while get into the bus and out from bus.
- Maintaining transparency in the community without any corruptions in the community.

8.2 Conclusion:

IOT based passenger detection and counting system has been proposed in this project. The passengers are detected under different challenging scenarios such as more number of passengers at a time. The proposed algorithm was tested using the existing dataset which provided good result. It was noted that the passengers who are moving toward or away from the sensor and changing dimensions of the passenger body size affects the accuracy of the detection system.

The proposed system can be extended to dynamic environments using advanced methods. The use of the proposed detection system can be extended in security application's such as activity recognition. Similarly, the counting system can be used for measuring economic activities such as in a shopping mall.

On the other hand, the Passenger Counting systems, can be much more appropriate and of greater interest, as related to this short analysis. It is important to underline that – even though most of them are still at a development stage – the passenger counting technologies available on the market are various and the issues of different kinds; the combinations of technologies are such that no solution can be considered to date better than others or economically preferable a priori; every solution should be analysed in detail for applying it thereafter to the actual conditions of the public transport system or company.

Program Outcomes (POs):

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs):

Engineering students will be able to

1. Process, interpret the real-world data to formulate the model for predicting and forecasting.
2. Apply machine learning techniques to design and develop automated systems to solve real world problems.

PROJECT PROFORMA

Classification of Project	Application	Product	Research	Review

Note: Tick Appropriate category

Project Outcomes	
Course Outcome(CO1)	Identify community needs/problems
Course Outcome(CO2)	Investigate different possible solutions to solve the problem
Course Outcome(CO3)	Make use of community involvement in solving the problem.
Course Outcome(CO4)	Prepare a report on the problem and its solution

Mapping-Table

Note: Map each project outcomes with POs and PSOs with either 1 or 2 or 3 based on level of mapping as follows:

1-Slightly (Low) mapped 2-Moderately (Medium) mapped 3-Sustainability (High) Mapped

AD3503: INTERNSHIP/INDUSTRIAL TRAINING/PRACTICAL TRAINING/ COMMUNITY PROJECT															
Course outcomes	Program Outcomes and Program Specific Outcome														
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS1	PS2	PS3
CO1: identify community needs/problems	1	3				2	1	1	2	2	2	1		2	2
CO2: Investigate different possible solutions to solve the problem		2	3			3	2	2	3	3	2	2		2	2
CO3: Make use of community involvement in solving the problem.			2			2	1	1	2	2	2			1	1
CO4: Prepare a report on the problem and its solution	2	1	2		2				2	2		2		2	2
INTERNSHIP/INDUSTRIAL TRAINING/PRACTICAL TRAINING/ COMMUNITY PROJECT	1	2	2		1	2	1	1	3	3	2	2		2	2

